27. Draw structures and provide IUPAC names for each of the following compounds.

a. Methyl ethyl ketone;

b. ethyl isobutyl ketone;

c. methyl *tert*-butyl ketone;

d. diisopropyl ketone;

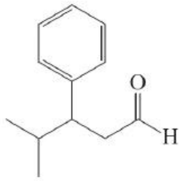
e. acetophenone;

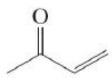
f. *m*-nitroacetophenone;

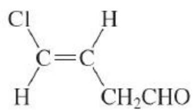
g. cyclohexyl methyl ketone.

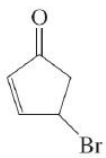
28. Name or draw the structure of each of the following molecules.

a. $(CH_3)_2CH\overset{\displaystyle O}{\overset{\|}{C}}CH(CH_3)_2$

b. 

c. 

d. 

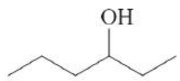
e. 

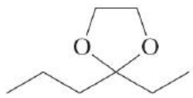
29. The following spectroscopic data are for two carbonyl compounds with the formula C8H12O.$C_8H_{12}O.$ Suggest a structure for each. The letter "m" stands for the appearance of this particular part of the spectrum as an uninterpretable multiplet. **(a)** 1H NMR: δ=1.60 (m, 4 H), ^1H NMR : $\delta = 1.60$ (m, 4 H), 2.15 (s, 3 H),2.15 (s, 3 H), 2.19 (m, 4 H), 2.19 (m, 4 H), 6.78 (t, 1 H) ppm;6.78 (t, 1 H) ppm; 13C NMR (DEPT): δ=21.6(CH2), ^{13}C NMR (DEPT) : $\delta = 21.6(CH_2)$, 22.0(CH2),$22.0(CH_2)$, 23.0(CH2),$23.0(CH_2)$, 25.1(CH3),$25.1(CH_3)$, 26.1(CH2),$26.1(CH_2)$, 139.7(Cquat),$139.7(C_{quat})$, 140.9(CH), 140.9(CH), 199.2(Cquat) ppm.$199.2(C_{quat})$ ppm. **(b)** 1H NMR: δ=0.94 (t, 3 H), ^1H NMR : $\delta = 0.94$ (t, 3 H), 1.48 (sex, 2 H),1.48 (sex, 2 H), 2.21 (q, 2 H),2.21 (q, 2 H), 5.8–7.1 (m, 4 H),5.8–7.1 (m, 4 H), 9.56 (d, 1 H) ppm;9.56 (d, 1 H) ppm; 13C NMR (DEPT): δ=13.6(CH3), ^{13}C NMR (DEPT) : $\delta = 13.6(CH_3)$, 21.9(CH2),$21.9(CH_2)$, 35.2(CH2),$35.2(CH_2)$, 129.0(CH),129.0(CH), 135.2(CH),135.2(CH), 146.7(CH),146.7(CH), 152.5(CH),152.5(CH), 193.2(CH) ppm.193.2(CH) ppm.

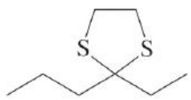
30. The compounds described in Problem 29 have very different ultraviolet spectra. One has $\lambda max(\varepsilon)=232(13,000)$ and $308(1450)$ nm $\lambda_{\max}(\acute{E}) = 232(13{,}000) \text{ and } 308(1450) \text{ nm}$, whereas the other has $\lambda max(\varepsilon)=272(35,000)$ nm $\lambda_{\max}(\acute{E}) = 272(35{,}000) \text{ nm}$ and a weaker absorption near 320 nm (this value is hard to determine accurately because of the intensity of the stronger absorption). Match the structures that you determined in Problem 29 to these UV spectral data. Explain

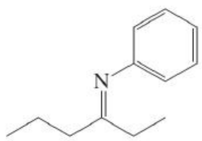
31. Following are spectroscopic and analytical characteristics for an unknown substance. Propose a structure. Empirical formula: C8H16O. $C_8H_{16}O$. 1H NMR: δ=0.90 (t, 3 H), $^1H\ NMR: \delta = 0.90$ (t, 3 H), 1.0–1.6 (m, 8 H),1.0–1.6 (m, 8 H), 2.05 (s, 3 H),2.05 (s, 3 H), 2.25 (t, 2 H) ppm;2.25 (t, 2 H) ppm; 13C NMR^{13}C NMR seven signals between δ=14.0$\delta = 14.0$ andand 43.8 ppm,43.8 ppm, and one at 208.9 ppm;208.9 ppm; IR: 1715 cm−1. 1715 cm^{-1}. UV: λmax(ε)=280(15) nm;UV : $\lambda_{max}(\acute{E}) = 280(15)$ nm; MS: m/z=128 (M+·);MS : $m/z = 128\ (M^{+\cdot})$; intensity of (M+1)+ $(M+1)^+$ peak is 9%9% of M+·$M^{+\cdot}$ peak; important fragments are at m/z=113 (M-15)+,$m/z = 113\ (M-15)^+$, m/z=85 (M-43)+, $m/z = 85\ (M-43)^+$, m/z=71 (M-57)+,$m/z = 71\ (M-57)^+$, m/z=58 (M-70)+$m/z = 58\ (M-70)^+$ (the second largest peak), and m/z=43 (M-85)+$m/z = 43\ (M-85)^+$ (the base peak).

32. Reaction review. Without consulting the Reaction Road Map on pp. 850–851, suggest reagents to convert each of the starting materials below into 3-hexanone.

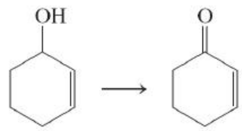
a. 

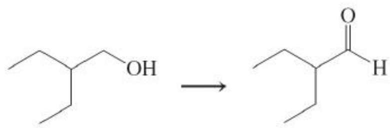
b. 

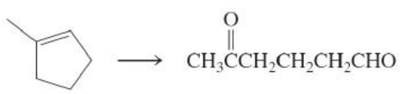
c. 

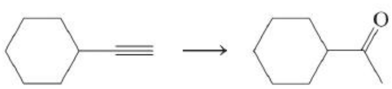
d. 

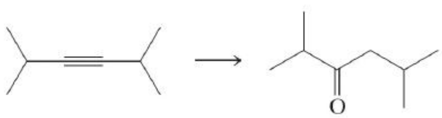
33. Indicate which reagent or combination of reagents is best suited for each of the following reactions.

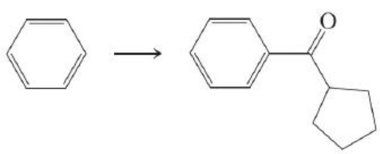
a. 

b. 

c. 

d. 

e. 

f. 

34. Write the expected products of ozonolysis (Section 12-12) of each of the following molecules.

35. For each of the following groups, rank the molecules in decreasing order of reactivity toward addition of a nucleophile to the most electrophilic sp2sp^2-hybridized carbon.

a. $(CH_3)_2C{=}O$, $(CH_3)_2C{=}NH$, $(CH_3)_2C{=}\overset{+}{O}H$

b. $CH_3\overset{\overset{O}{\|}}{C}CH_3$, $CH_3\overset{\overset{O}{\|}}{C}\overset{\overset{O}{\|}}{C}CH_3$, $CH_3\overset{\overset{O}{\|}}{C}\overset{\overset{O}{\|}}{C}\overset{\overset{O}{\|}}{C}CH_3$

c. BrCH2COCH3,×CH3COCH3,×CH3CHO,×BrCH2CHO
$BrCH_2COCH_3$, CH_3COCH_3, CH_3CHO, $BrCH_2CHO$

36. Give the expected products of reaction of butanal with each of the following reagents.

a. LiAlH4, (CH3CH2)2O, $\mathrm{LiAlH_4,\ (CH_3CH_2)_2O,}$ then H+, H2O $\mathrm{H^+,\ H_2O}$

b. CH3CH2MgBr, (CH3CH2)2O, $\mathrm{CH_3CH_2MgBr,\ (CH_3CH_2)_2O,}$ then H+, H2O $\mathrm{H^+,\ H_2O}$

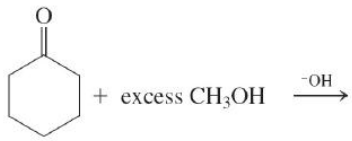
c. HOCH2CH2OH, H+ $\mathrm{HOCH_2CH_2OH,\ H^+}$

37. Give the expected products of reaction of 2-pentanone with each of the reagents in Problem 36.

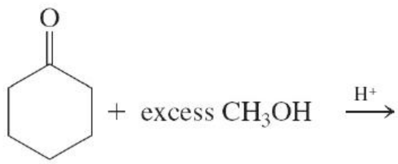
38. Give the expected products of reaction of 1-(3-cyclohexenyl)-1-

39. Give the expected product(s) of each of the following reactions.

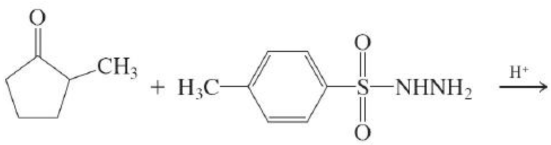
a.



b.



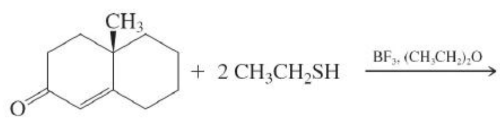
c.



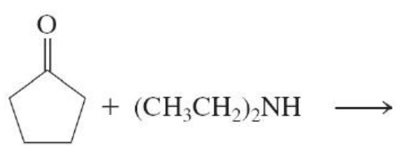
d.

$$CH_3\overset{\overset{\displaystyle O}{\|}}{C}CH_3 + HOCH_2\overset{\overset{\displaystyle OH}{|}}{C}HCH_2CH_2CH_3 \xrightarrow{H^+}$$

e.



f.



40. The equations for the mechanism of hydration of acetaldehyde under acidic aqueous conditions below are flawed. **(a)** For each step, add the missing electron pairs and identify the problem(s). **(b)** Write a correct step-by-step mechanistic description of the process.